

→ field $\psi(x)$

2. Lagrangian formalism

a) problem set-up:

1) Let's introduce some field φ which depends on a space-time point.

$$x = (x^0 = ct, x^i) \equiv (x^m), \quad 0 \leq m \leq D-1,$$

total dimension of space-time ($D = 4$).

x belongs to some space, $x \in \Omega_D \subset \mathbb{R}^D$.

ex : with the elastic string of length L , $D = 2$,

$$\Omega_2 = [t_1, t_2] \times [0, L].$$

More generally, all definition domains will look like

$$\Omega_D = [t_1, t_2] \times \Omega_{D-1},$$

spatial part.

2) Dynamics: we have a Lagrangian density

$$\mathcal{L}(\varphi, \partial_m \varphi; x), \quad \partial_m \varphi = \frac{\partial \varphi}{\partial x^m}.$$

3) The action then reads:

$$S = \int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x) = \int_{t_1}^{t_2} dt \left[\int_{\Omega_{D-1}} d^{D-1} x \mathcal{L}(\varphi, \partial_m \varphi; x) \right]$$

Lagrangian L

point-like $\alpha = 1$

• b) equation(s) of motion:

We will use the same type of reasoning:

- Introduce a small variation of the field.
- Impose that S is stationary ($\delta S = 0$).

Starting point:

$$\begin{aligned} S &= \int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x). \\ \begin{cases} \varphi(x) \longrightarrow \varphi(x) + \delta \varphi(x), \\ \partial_m \varphi \longrightarrow \partial_m \varphi + \partial_m \delta \varphi(x). \end{cases} \\ S' &= \int_{\Omega_D} d^D x \mathcal{L}(\varphi + \delta \varphi; \partial_m \varphi + \partial_m \delta \varphi; x) \\ &= \int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x) \\ &\quad + \int_{\Omega_D} d^D x \left[\frac{\partial \mathcal{L}}{\partial \varphi} \delta \varphi + \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \partial_m \delta \varphi \right] + \text{higher orders.} \end{aligned}$$

δS Einstein convention: summation over repeated indices implied

However:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \partial_m \delta \varphi &= \partial_m \left(\frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \delta \varphi \right) - \left(\partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right) \delta \varphi \\ \partial_\mu A^\mu &= \partial_m A^m \\ (u v') &= (uv)' - u'v. \end{aligned}$$

Then:

$$\delta S = \int_{\Omega_D} d^D x \left[\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right] \delta \varphi + \left[\int_{\Omega_D} d^D x \partial_m \left(\frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \delta \varphi \right) \right]$$

divergence integral

Assumption: the boundary condition on $\delta\varphi(x)$ implies that the divergence term vanishes

$$\delta\varphi(x) = 0, \quad \forall x \in \text{surface of } \Omega_D.$$

Hence, $\forall \delta\varphi(x)$,

$$\delta S = \int_{\Omega_D} d^D x \left[\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right] \delta\varphi.$$

We impose $\delta S = 0$ because of the $\forall \delta\varphi$:

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} = 0$$

Euler-Lagrange equation of field.

Comment: $\mathcal{L}$ and $\mathcal{L} + \partial_m k^m$ give the same equation of motion.

• c) notion of functional:

T is a functional $\iff$ it is a function of a function.

T : a function space $\longrightarrow \mathbb{R}$ or $\mathbb{C}$

$$f \longmapsto T[f] \quad \textcolor{red}{functional}[]$$

We can say that a functional is the generalization of a function which depends on a set of different variables

$$f(x^1, x^2, \dots)$$

to a continuous set of variables.

$$\text{ex.} \quad S[\varphi] \quad S'[\varphi + \delta\varphi]$$

Remark: derivative of a functional.

in the context of a multi-variable function:

$$f(x^1, x^2, \dots) \quad \textbf{Pb:} \quad \frac{\partial f}{\partial x^i} ??$$

$$\frac{\partial f}{\partial x^i} = \lim_{\epsilon \rightarrow 0} \frac{f(x^1, x^2, \dots, x^i + \epsilon, \dots) - f(x^1, x^2, \dots, x^i, \dots)}{\epsilon}$$

$$\text{notation} = \lim_{\epsilon \rightarrow 0} \frac{f(x^j + \epsilon \delta_i^j) - f(x^j)}{\epsilon}, \quad \delta_i^j = \text{Kronecker delta.}$$

★ functional case:

$$f \longmapsto T \quad x^j \longmapsto \varphi(y)$$

Hence:

$$\frac{\delta T}{\delta \varphi(x)} = \lim_{\epsilon \rightarrow 0} \frac{T[\varphi(y) + \epsilon \delta^D(x - y)] - T[\varphi(y)]}{\epsilon}$$

$$\delta^D(x) = D\text{-dimensional Dirac "function"}$$

Note: remember that

$$\delta S = \int_{\Omega_D} d^D x \left(\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right) \delta\varphi$$

$$\stackrel{\text{definition}}{=} \int_{\Omega_D} d^D x \frac{\delta S}{\delta \varphi(x)} \delta\varphi(x)$$

$$\frac{\delta S}{\delta \varphi(x)} = \frac{\delta S}{\delta \varphi(x)} = \frac{\partial \mathcal{L}}{\partial \varphi(x)} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi(x))}.$$

Exercise

Compute $\frac{\delta T}{\delta \varphi(x)}$.

For

- $T[\varphi] = \frac{1}{2} \int_{\Omega_D} d^D x \varphi(x)^2$
- $T[\varphi] = \varphi(y)$
- $T[\varphi] = \frac{1}{2} \int_{\Omega_D} d^D x d^D y \varphi(x) G(x, y) \varphi(y)$

$$\text{with} \quad G(x, y) = G(y, x).$$

→ Hamiltonian formalism

• a) discrete case:

Let's go back to the elastic string.

Starting point: Lagrangian

$$L = a \sum_i L_i = a \sum_i \left(\frac{1}{2} \frac{M}{a} \dot{\varphi}_i^2 - \frac{1}{2} k a \left(\frac{\varphi_{i+1} - \varphi_i}{a} \right)^2 \right).$$

By definition, the conjugated momentum of φ_i is

$$\Pi_i \stackrel{\text{def}}{=} \frac{\partial L}{\partial \dot{\varphi}_i} = a \frac{\partial L_i}{\partial \dot{\varphi}_i}.$$

Then the Hamiltonian is:

$$H = \left(\sum_i \Pi_i \dot{\varphi}_i \right) - L = a \sum_i \left(\frac{\Pi_i}{a} \dot{\varphi}_i - L_i \right).$$

• b) continuous limit:

$$L_i \longrightarrow \mathcal{L}, \quad L \longrightarrow \int_{\Omega_{D-1}} d^{D-1}x \mathcal{L}, \quad a \sum_i \longrightarrow \int_{\Omega_{D-1}} d^{D-1}x \quad (\text{space-only}).$$

$$\frac{\Pi_i}{a} = \frac{\partial L_i}{\partial \dot{\varphi}_i} \longrightarrow \boxed{\Pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}}} \quad \left(\partial_0 \equiv \frac{\partial}{\partial t} \right)$$

Conjugate momentum density (FIELD)

Finally

$$H = \int_{\Omega_{D-1}} d^{D-1}x [\Pi(x) \dot{\varphi}(x) - \mathcal{L}(\varphi, \partial_m \varphi; x)]$$

Hamiltonian density $\mathcal{H}$ **Comment:**

- Lagrangian $\longrightarrow S$ (time & space are on the same foot)
- Hamiltonian $\longrightarrow H$ (time is singled out)

→ Symmetries and Noether theorem

• a) Problem set-up:

Let's start with a field $\varphi(x)$ whose dynamics is driven by

$$\mathcal{L}(\varphi, \partial_m \varphi; x).$$

$$S[\varphi] = \int_{\Omega_D} d^Dx \mathcal{L}(\varphi, \partial_m \varphi; x).$$

There are two types of transformations which can modify the fields:

① Continuous transformation:

They can be infinitesimal, local, non-local

$$\text{ex. } t \longrightarrow t' = t + a$$

$$|a| \ll 1 \quad \text{infinitesimal transf.}$$

$$a = ct \quad \text{non-local transf.}$$

$$a(x) \quad \text{local transf.}$$

Note: Transformations which act upon the coordinates are geometrical transf.

Transformations which act upon the field density are called internal transf.

② Discrete transformations

Those ones cannot be infinitesimal.

$$\text{ex. } t \longrightarrow -t$$

From now on, we will use continuous transformations, i.e. we will use a transformation which can act on the field and its coordinates:

$$\begin{aligned}x^M &\longrightarrow x'^M = x^M + \delta x^M, & \delta x^M \ll 1 \\ \partial_M &= \frac{\partial}{\partial x^M}, & \varphi(x) \longrightarrow \varphi'(x') \\ \partial_M \varphi(x) &\longrightarrow \partial'_M \varphi'(x').\end{aligned}$$

• b) Symmetry

Definition: the transformation is a symmetry

$$\Longleftrightarrow \text{the action after the transformation is equal to the action before} \Longleftrightarrow S[\varphi] = S'[\varphi']$$

$$\Longleftrightarrow \underbrace{\int_{\Omega'_D} d^D x' \mathcal{L}'(\varphi', \partial'_m \varphi'; x')}_{S'[\varphi']} = \underbrace{\int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x)}_{S[\varphi]}.$$

Lagrangian density after the transformation

Remark: a symmetry gives another possible motion of a system.

ASSUMPTION $\longrightarrow$ the invariance of the action may result from different relationships between $\mathcal{L}$ and $\mathcal{L}'$.

The two most natural possibilities are:

1. both Lagrangian have the same mathematical structure

$$\mathcal{L}'(\varphi', \partial'_m \varphi'; x') = \mathcal{L}(\varphi', \partial'_m \varphi'; x')$$

2. They differ by a divergent term

$$\mathcal{L}'(\varphi', \partial'_m \varphi'; x') = \mathcal{L}(\varphi', \partial'_m \varphi'; x') + \partial'_m K^m(\varphi', \partial'_m \varphi'; x').$$

However, for all the transformations encountered in these lectures and almost in real life (no counter example found yet); the #2 case never occurs.

CONSEQUENCE

$$\begin{aligned}S'[\varphi'] &= S[\varphi] \\ \Longleftrightarrow \int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x) &= \int_{\Omega'_D} d^D x' \mathcal{L}'(\varphi', \partial'_m \varphi'; x') \\ \text{change of variable } x' \rightarrow x &\implies = \int_{\Omega_D} d^D x \left| \frac{\partial x'^\mu}{\partial x^\nu} \right| \mathcal{L}'(\varphi', \partial'_m \varphi'; x).\end{aligned}$$

Determinant of Jacobian matrix here, functions of $\varphi, \partial_m \varphi, x$

By identification,

$$\begin{aligned}\mathcal{L}(\varphi, \partial_m \varphi; x) &= \left| \frac{\partial x'^\mu}{\partial x^\nu} \right| \mathcal{L}'(\varphi', \partial'_m \varphi'; x) \\ &= \left| \frac{\partial x'^\mu}{\partial x^\nu} \right| \mathcal{L}(\varphi', \partial'_m \varphi'; x)\end{aligned}$$

Using the assumption on $\mathcal{L}$ and $\mathcal{L}'$

Pb Perform a Taylor expansion at first order of $S[\varphi] = S'[\varphi']$ to get the Noether theorem.